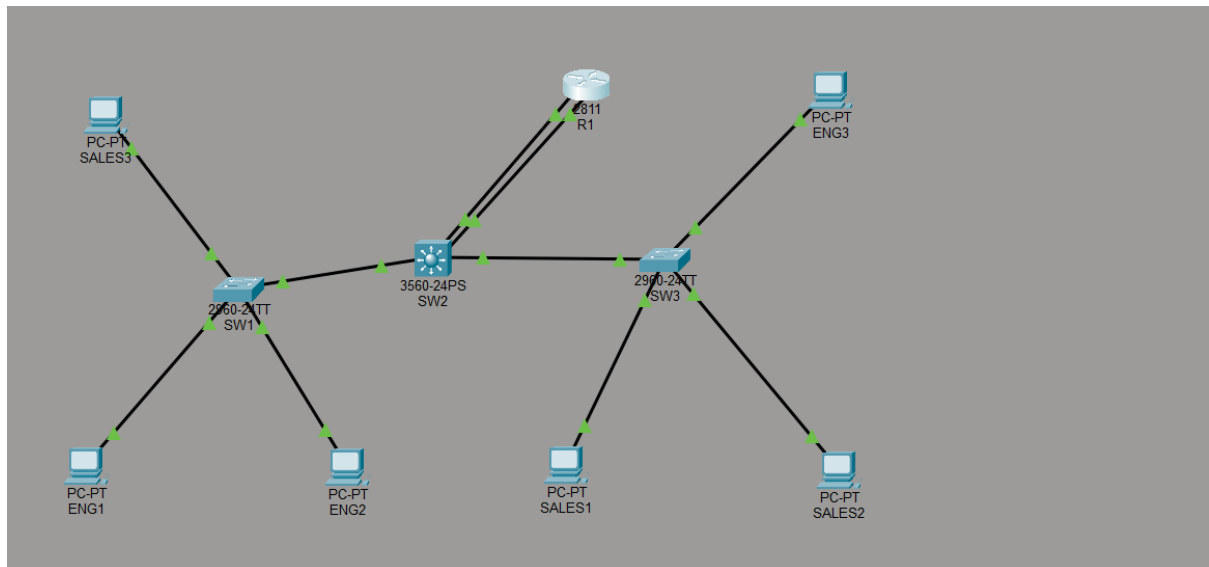


Image of topology.



VTP, Access and Trunk Ports

89. Configure VTP on the switches : choose a domain name, mode, and password, and document them in your submission .

```
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain CampusNet
Changing VTP domain name from NULL to CampusNet
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
CampusNet.

Switch(config)#vtp mode server
Device mode already VTP SERVER.
Switch(config)#vtp password Cisco123
Setting device VLAN database password to Cisco123
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Switch 1 configurations complete above

```

Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain CampusNet
Changing VTP domain name from NULL to CampusNet
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
CampusNet.

Switch(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch(config)#vtp password Cisco123
Setting device VLAN database password to Cisco123
Switch(config)#

```

Switch 2 configurations complete above

```

Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vtp domain CampusNet
Changing VTP domain name from NULL to CampusNet
Switch(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
CampusNet.

Switch(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch(config)#vtp password Cisco123
Setting device VLAN database password to Cisco123
Switch(config)#

```

Switch 3 configurations complete above

Showing vtp status

```

Switch#show vtp status
VTP Version capable      : 1 to 2
VTP version running     : 1
VTP Domain Name          : CampusNet
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 00D0.FF5A.8B00
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
Configuration Revision   : 0
MD5 digest               : 0x7E 0xAF 0x2E 0x8C 0xF8 0x56 0xB8 0x60
                        : 0xA6 0xC1 0x05 0xF9 0x1D 0xBF 0x03 0xD6

```

90. Create the VLANs required by the topology and assign access ports to the correct VLANs.

```
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name engineering
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name sales
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Vlan created and assigned for vlan 10

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Vlan created and assigned for vlan 20

91. Configure Trunk Ports

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up

Switch(config-if)#switchport trunk allowed 10,20
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#end
Switch#
%SYS-5-CONFIG I: Configured from console by console
```

92. Verify VLAN and Trunk Configuration

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	engineering	active	Fa0/2
20	sales	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

```
Port Vlan
```

Port	Vlans allowed on trunk
Gig0/1	10,20

```
Port Vlan
```

Port	Vlans allowed and active in management domain
Gig0/1	10,20

```
Port Vlan
```

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	10,20

```
Switch#
```

93.Inter-VLAN Routing (Option 1: Separate Interfaces)

Press RETURN to get started!

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface fa0/1
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
```

Configuration on router 1

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuration on switch 2

94. Inter-VLAN Routing (Option 2: Router on a Stick)

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#no shutdown
Router(config-if)#interface fa0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed
state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.10.10.1 255.255.255.0
% 10.10.10.0 overlaps with FastEthernet0/0
Router(config-subif)#interface fa0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed
state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
% 10.10.20.0 overlaps with FastEthernet0/1
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```